

Free education

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Google Maps 6.0 now indoors

Google Maps now includes indoor mapping and positioning inside select malls, airports etc.
<http://dgit.in/18jhiuy>

Power Banks Feature

because once in a blue moon you might need a power fix. So for an everyday carry device, it has to be as light as possible. Given that weight is directly proportional to the capacity of a power bank you might have to compromise on capacity.

Volume and form factor

Though volume isn't a factor which you'd give a lot of emphasis to, it's something you can't ignore completely. Compact form factors are preferred because they don't cause any hindrance when you want to use your device while it's being charged.

One for all

A powerbank that has the ability to charge every device that you own is one of the most important factors. Having to carry one power bank for your phone and another for your tablet is too much of a fuss. Then there is the issue of having many interfaces to comply with, though chargers these days don't have the cable affixed to it which means that you can make do with just a USB port on your power bank but for older generation devices it makes sense to have adaptors. Hence, a power bank that can not only charge all your devices but also do it simultaneously would be the perfect choice.

Where are the scores?

Once you are through with the entire article you're likely going to figure out that this isn't a proper test that we at Digit do. Well there are quite a few reasons as to why it turned out so and we've elaborated upon them below.

Since we're basically testing batteries the aspects we wanted to focus on included the rate of

charge and rate of discharge. The battery indicators provide a rough approximation and there was no way approximate figures could be used to compare the performance between two products, that's something we don't like to compromise on. So in order to combat this we needed a accurate battery tester. Ripples in the voltage is also a factor we had to find a way to deal with, at the interface between the actual device and its battery there can only be about 1% variance around 4.2volts, however, this is handled by the device hardware. What we wanted to check is if the voltage to the device was within acceptable limits. A multimeter does give a good read but ripples will only be evident with an oscilloscope sporting a decent sampling rate. We didn't have either equipment but managed to find some for rent.

Then we wanted to find out which brand of batteries the manufacturers were using. They could have been cheap rip-off batteries, OEM batteries or some of the best that the market has to offer. Alas, this meant that we needed to break the case open to check out the insides. What went wrong is that not all manufacturers were fine with their products being torn apart so in order to maintain a standard treatment to each brand we had to chuck the entire testing process altogether.



CyberPower CPBS10400 - can charge an iPad completely



Cooler Master PowerFort 6600 - good build quality

Now comes the time part. Manufacturers use ideal situations and end up with specifications a bit further than the truth. This is why a standardised discharge and charge test was needed. Also we needed to test the same battery at different loads which meant that we'd have to charge and discharge each power bank twice or thrice for each test. Each powerbank would take anywhere from 4-12 hours to charge and the high capacity ones needed an entire day. So effectively we required 2 days on an average per power bank. With 28 power banks this meant that we needed to work round the clock for 56 days which was far from possible given the time frame that we had to work with. However, since there were a lot of our readers asking for this comparison we thought this was an apt way to get the information across to you without going into all the technical mimbo jumbo.

Some might wonder, why not just charge and discharge using a standard device like an iPad with its massive battery? It's not that simple, modern smartphones and tablets don't have a linear charge indicator, i.e., if the battery is at ten percent capacity then the device might read four percent. Likewise, when a device ranges

between 70-90 percent it might only indicate 60-70 percent charge. Which is why your battery seems to lose the initial charge rather quickly and lasts ages on the final 10% charge. To top it off some devices show that they have 100% charge prematurely and continue charging in the background. This non-linearity means that we would never be able to accurately measure across more than one charge cycle.

Features

Indicator lights

All power banks that we looked at had charge level indicators in some form or the other. Some would have just the one LED which would change colours if they were low on charge. The most common arrangement was to have 4-5 LEDs which would all light up when the power bank was at full charge and then one by one the LEDs would either turn off or change colours to indicate a drop in charge level. Thus, each LED stands for 0 to 20-25% charge. While this may be more than enough for the average consumer, power users and "geeks" prefer a much accurate scale which only one power bank could provide for - the AsiaPower AP-5200A. This unit has a two-digit LED counter which is placed beneath the dark tinted plastic body. Being the only one with the particular feature coupled with a minimalist design, it did look clean and polished.

Amperage range

Having more than one output is definitely a huge plus, but the important part is that each output must be able to supply the full amperage that it is rated for, i.e., the current should not be divided across the two